

PROJECT: ANALYSING FITBIT DATA

Introduction

In this project, we study a dataset with information obtained from fitbits generated by 33 respondents to an amazon survey in 2016 that submitted censored usage data. The end product will be a dashboard that can be used to study various statistical analysis of the users.

Before starting, please have a careful look at the rules for the [use of ChatGPT and other tools for generative AI](#) and note that you are required to keep a log of your usage of GenAI.

Part 1: Getting acquainted with the data

In the first part, we will just work with the file `daily_activity.csv` that consists of various statistics that the participants of the survey reported daily. In the first part of the assignment, you will explore this dataset and make some basic computations and visualizations.

Basic inspection of the data

Count how many unique users there are in the dataset and then compute the total distance the fitbits registered for each user. Display the results in a suitable graph.

To make it more user-specific, implement a function that takes as input a user Id and displays a line graph that shows the calories burnt on each day. Improve your function to select a specific range of dates in which results are shown. You may want to use the library `datetime` for this purpose.

Use `datetime` or manually look up the day of the week for each date and make a barplot with the day on the x -axis and the frequency at which all individuals work out on the respective day on the y -axis.

Relationship between calories and steps taken

Based on the amount of steps taken, we would like to estimate the calories burnt based on the available data. Next we investigate the relationship between the amount of steps taken and the amount of calories burnt. Use the function `OLS` from `statsmodels` to set up a the following linear regression model

$$\text{Calories} = \beta_0 + \beta_1 \cdot \text{TotalSteps} + \beta_2 \cdot \text{Id} + \epsilon$$

Where $\epsilon \sim N(0, \sigma^2)$. Here, `Id` is taken as explanatory variable as the relationship could possible be different for each individual. Make sure that `Id` is interpreted as a factor here.

Next write a function that take `Id` as input and produces a figure in which the linear relationship for that `Id` is visualized in a scatterplot + regression line.

Use your own creativity to discover any other features about this dataset you find interesting or noteworthy! Look for relations between variables of create attractive visualizations of the data. Points will be awarded for creative insights or features of the final product!

Create one `.py`-file which, upon execution produces all required figures

Part 2: Setting up a GitHub repository

The final product needs proper version control. Therefore everyone in the group will need an account on [GitHub](#). Set up a private repository with your group and add a `.py`-file to it containing the code used for Part 1. Make sure to add a `README.md` file to it.

The `README.md`-file is very important as it contains information for users of the software you create. It should therefore serve as a guide for a use to use your library. `.md`-files can be opened and edited in any text editor, including Visual Studio Code.

Initialize `git` Take turns within your group and add small sections to the `README.md`-file each time. Between these turns `commit` and `push` the changes until you are satisfied with the file. Make sure to keep updating the `README` throughout the project, as this file will also be judged at the end.

Continue working on Part 1, while continuously using the GitHub workflow. Also consider adding extra `.py`-scripts for specific parts of the assignment.

Part 3: Interacting with the database

We classify the participants of the survey in three classes:

- Light user: If the `Id` appears 10 times or less in `daily_activity`.
- Moderate user: If the `Id` appears 11 to 15 times in `daily_activity`.
- Heavy user: If the `Id` appears 16 times or more in `daily_activity`.

Create a new `pd.DataFrame` that contains the columns `Id` and `Class` and contains for each unique `Id` the `Class` to which it belongs.

The file `fitbit_database.db` is now available on Canvas. For the remainder of the project, you have to add the database to your GitHub repository.

It could be that the database is too large to upload to GitHub directly through the website. This is because there is a size limit of 25MB here. It is possible to add them to the repository via the command line (the limit here is 100 MB). A detailed instruction as how to do this is on [this web page](#) in the section “Adding a file to a repository using the command line”.

This database contains the following (tabular) datasets.

- `daily_activity`: The file you have already been working with.
- `heart_rate`: Heart rate of each individual measured every 5 seconds.
- `hourly_calories`: Calories burnt per hour.
- `hourly_intensity`: Intensity of exercise, given both in total per hour and average per hour.
- `hourly_steps`: Total amount of steps taken per hour
- `minute_sleep`: Information on every minute that the participant is asleep.
- `weight_log`: Information on the weight, fat, BMI for each of the participants.

Use the library `sqlite3` to connect to the database. It may be useful to first have a look at the variable names for each table first. However, for each task of the remaining parts, it is most prudent to write SQL-queries to access the parts of the database needed for that task, rather than loading it all into a few `pd.DataFrame`s.

For this part, you have to investigate the entire database and link the different tabular data together. There is a lot to investigate. Below are some interesting tasks to complete with this database, but as with part 1, feel free to investigate further!

- The table `minute_sleep` contains the variable `logId`, which assigns an id for each time. Compute the sleep duration for each moment of sleep of an individual.
- Analyse whether the duration of sleep is related to the active minutes for an individual. To do this, compute the total active minutes as the sum of the very active, fairly active and light active minutes on a day and perform a regression against the minutes of sleep on that day.

- Also analyse the relationship between the amount of sedentary activity and the sleep duration for all individuals by performing a linear regression on all data with the sleep duration as response variable and the sedentary activity as explanatory variables. Additionally, think of a way to visually verify that the model assumption that all errors are normally distributed can be verified.
- Divide the day in blocks of 4-hours; 0–4, 4–8, 8–12, 12–16, 16–20 and 20–24. Compute for each block how much steps are taken on average over all participants in that block and visualize the results in a barplot. Do the same for the calories burnt and minutes of sleep.
- Write a function that takes an `Id` as input and returns a figure that contains heart rate of this individual and the total intensity of the exercise taken.
- It is given that all participants of this survey live in Chicago. Information on the weather is widely available online. A good source for this is the website <https://www.visualcrossing.com/weather-query-builder/>. Write either a query to the webAPI or download the data concerning the dates of the fitbit data. Investigate if there is a relation between weather factors such as precipitation and temperature and the activity of the individuals in the database and make proper visualizations for the (non)-linear relationships found.